Fire Drill

UGREEN HACKATHON 2026 ON AI NAS · EXPLORE YOUR IMAGINATION (OPEN INNOVATION) TRACK

1. What the Skill does

Fire Drill finds the family photos that exist only once on your NAS, copies just that set to a folder you pick, and tells you when backups silently stopped or files vanished.

2. Target user and problem

Families who free phone space once photos reach the NAS. The NAS becomes the only copy, and RAID does not cover theft, fire, ransomware or a failed pool. Phone uploads stop silently in the background, and 46% of backup users never tested a restore (Macrium, 2017). On Feb 7, 2026 an AI agent tidying a desktop deleted 15,000 family photos.

3. Proposed solution

A drill the NAS runs on itself, ranked by what you could never replace. It reports, most urgent first: files vanished since the last drill, irreplaceable photos with one copy, silently stopped backups, broken or partial uploads, then a grade A to F per person with at most three fixes. Then it copies only the one-copy set.

4. Core user flow

"Run a fire drill" → confirm which folders are whose backups → evidence gathered → Report Card with receipts → confirm a destination → copy → confirm where to save the report. Later drills compare with the manifest and copy only new files.

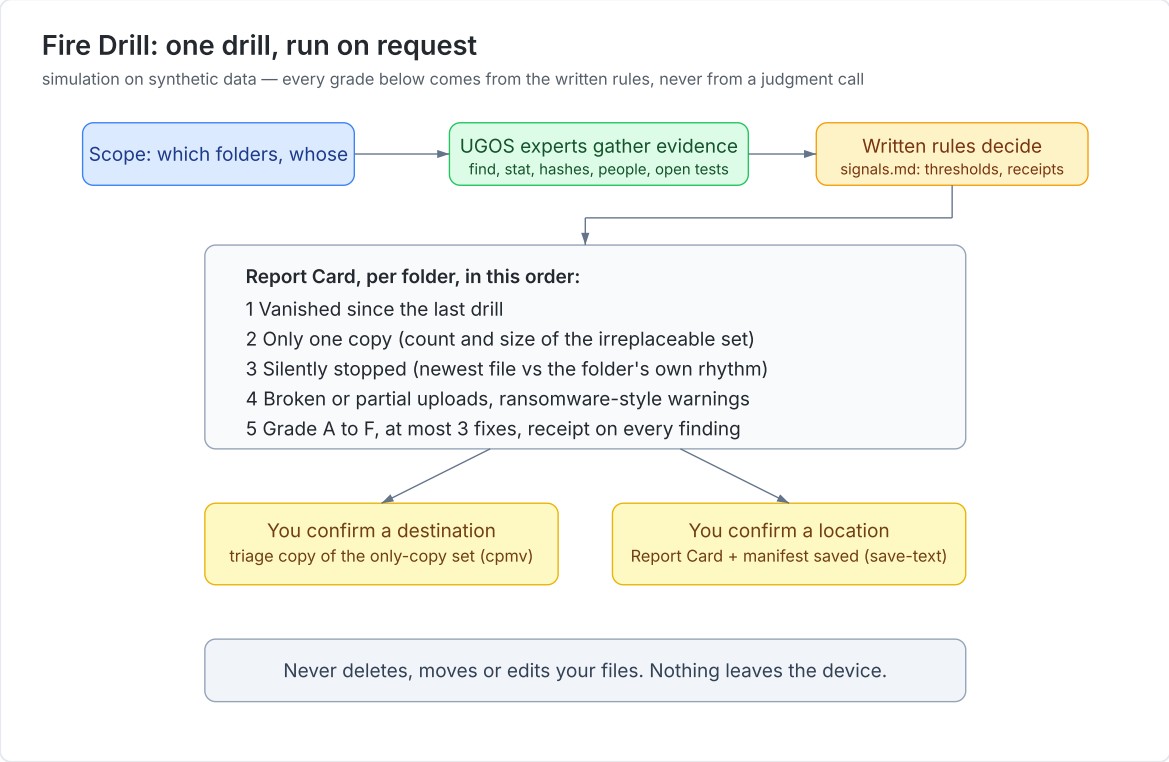


Figure 1. One drill: experts gather evidence, written rules decide, two confirmation gates.

5. Difference from existing tools

Backup tools check job status or hashes; neither proves a photo still opens, nor tells a daughter's birthday from a screenshot. Fire Drill uses on-device people search and vision to protect what matters, and turns the report into a confirmed fix small enough for a USB drive.

6. NAS capabilities by step

Scope and rhythm: ug-filesystem find, stat. Copy count: sha256sum, ug-filemgr scan-dedupe. Tiering: ug-photo search-picture, search-screen. Open test: vision-ocr, vision-vlm; ug-document qa, extract. Fix: ug-filesystem cpmv. Report and manifest: ug-knowledge save-text. All confirmed in the kit catalog.

7. Technical approach

A Markdown-only Skill. Written rules decide every flag and grade; experts only gather evidence. Inputs: folder scope and owners. Permissions: read-only experts need no approval; cpmv (copy, overwrite off) and save-text run only after user confirmation and UGOS approval. Models: UGOS on-device vision and OCR only. Data never leaves the NAS.

8. Development status

Implemented: Skill package (kit checker 0 errors, 0 warnings), reference simulator, 39 tests. Design pack 20/20 with 4 healthy controls unflagged. Adversarial holdout: 1/6 before fixes, 6/6 after; a second holdout scores 2/4. Pending: NAS execution.

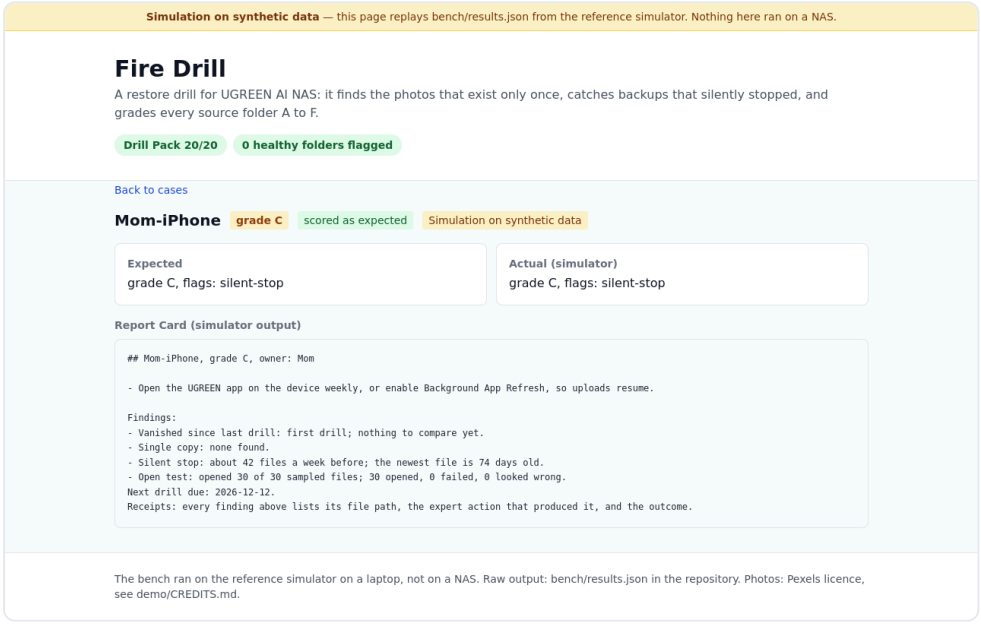


Figure 2. A Report Card from the reference simulator (synthetic data, not a NAS).

9. Validation plan

Core 1: a phone folder whose photos stopped 74 days ago while screenshots keep arriving is flagged with its last-photo date. Core 2: a one-copy family photo set is sized, copied after confirmation, and verified by hash. Edge: an empty readable folder and an unreadable one are reported differently, neither graded.

10. Known limitations

Sampling, not a full scan. Ransomware signs are warnings; in-place encryption of under 100 files can evade them. Seasonal folders must be marked by the user. UGOS lacks USB listing, snapshots and a scheduler, so the copy targets a folder you pick and drills run on request with a reminder.

Third-party components and data flows

(excluded from the word limit)

- **Skill package (the submitted ZIP):** no third-party components. Markdown only (SKILL.md and references), written for this entry.
- **UGOS capabilities:** used through the UGREEN skill kit (uliya-skill-kit 0.7.3), which is licensed by UGREEN and not redistributed; CI downloads it from the official starter-kit page.
- **Repository tooling, not in the package:** TypeScript (Apache-2.0); ESLint, @eslint/js, typescript-eslint, @types/node (MIT).
- **Demo app, not in the package:** React, React DOM, Vite, @vitejs/plugin-react, Tailwind CSS, PostCSS, Autoprefixer, @types/react (MIT). Three stock photos from Pexels (photos 1648387, 1128318, 1683975) under the Pexels licence; no real customer data.
- **External services and data flows:** none for the Skill; all evidence and results stay on the NAS. GitHub hosts the code and the static demo, which uses synthetic data only.

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